

Felipe Papa Capalbo

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in felipepcapalbo 📺 FelipePCapalbo

Summary

Production engineer specialized in data analysis and pragmatic problem-solving, with expertise in Digital Twin applications, simulation, and optimization of manufacturing and logistics processes. Routinely employs predictive and explanatory data analytics, leveraging machine learning and advanced statistical methods.

Experience

Business Analyst

FlexSim Brasil

Campinas, BR

Jul 2024 – Present

- Optimized preventive maintenance scheduling for Brazil's largest mining company, improving asset availability and increasing operational efficiency by 12%.
- Developed a causal demand forecasting model for a leading cosmetics brand using GBDT and GANs, integrating market data, holiday calendars, and sales history into SAP.
- Built a custom multi-agent LLM chatbot with RAG to support FlexSim modelers, streamlining simulation development and onboarding.
- Designed a metamodeling-based optimization framework for simulation models, reducing runtime and enhancing solution quality.
- Led a warehouse headcount optimization project using simulation and mathematical modeling, achieving a 16% performance improvement over baseline.

Intern Business Analyst

FlexSim Brasil

Campinas, BR

Aug 2023 – Jul 2024

- Designed and delivered training sessions on Autonomous Digital Twins using Reinforcement Learning applied to discrete event simulation environments.
- Acted as a data scientist in a Digital Twin project for an iron ore mining site in inland Brazil, contributing to data-driven simulation initiatives.

Prototype Development Engineer

Prototype Development Centre - UFSCar

São Carlos, BR

Oct 2022 – Present

- Coordinated the development of an IoT device for citrus harvest tracking and weighing, including mechanical prototyping and the integration of management systems with SQL and NoSQL databases.
(Register number: BR10202401124)

Education

UFSCar - Universidade Federal de São Carlos

Master of Science in Production Engineering

Mar 2025 – Present

- Thesis: Optimization of Latex Goods Production: Lot Sizing and Scheduling approach.

UFSCar - Universidade Federal de São Carlos

Bachelor of Science in Production Engineering

Mar 2020 – Feb 2025

- Final Paper: Autonomous Digital Twin protocol applied to cotton blending in textile spinning, with Reinforcement Learning and Discrete Event Simulation ([PDF](#) [↗](#))

Skills

Programming Languages: Python, C, Java, VBA, JavaScript, SQL, MongoDB.

Technologies: Digital Twin, Statistical Analysis, Machine Learning and Mathematical Modelling.

Spoken Languages: Portuguese (native), English (C1), Spanish (A2), Mandarin (A2).